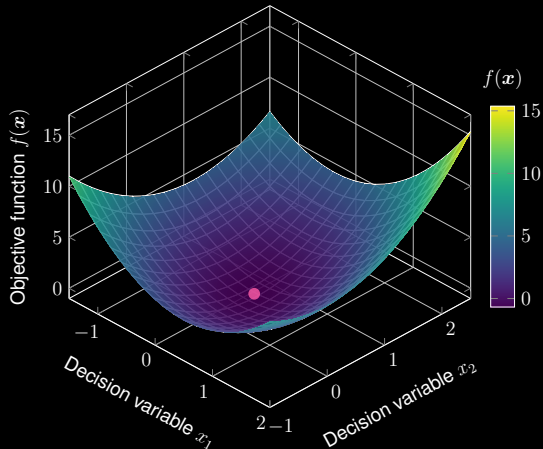
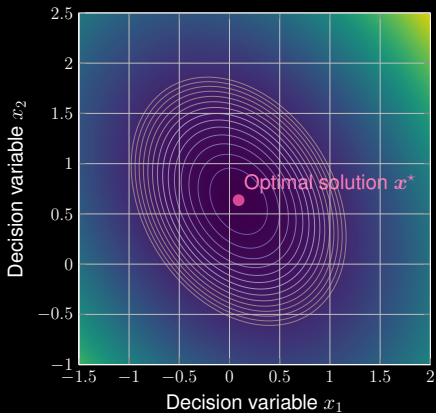


Quadratic Optimization

$$\min_{\mathbf{x} \in \mathbb{R}^2} \frac{1}{2} \mathbf{x}^\top \mathbf{A} \mathbf{x} - \mathbf{b}^\top \mathbf{x} \quad \text{with} \quad \mathbf{A} = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$



Steepest Gradient Descent for Quadratic Optimization

For any symmetric positive definite matrix A and vector b :

$$\min_{x \in \mathbb{R}^2} f(x) = \frac{1}{2} x^\top A x - b^\top x$$

with an initial point x_0 .

Steepest Gradient Descent for Quadratic Optimization

For any symmetric positive definite matrix A and vector b :

$$\min_{x \in \mathbb{R}^2} f(x) = \frac{1}{2} x^\top A x - b^\top x$$

with an initial point x_0 .

❶ Gradient (first-order derivative of $f(x_k)$):

$$g_k = A x_k - b$$

Steepest Gradient Descent for Quadratic Optimization

For any symmetric positive definite matrix A and vector b :

$$\min_{x \in \mathbb{R}^2} f(x) = \frac{1}{2} x^\top A x - b^\top x$$

with an initial point x_0 .

❶ Gradient (first-order derivative of $f(x_k)$):

$$g_k = A x_k - b$$

❷ Decision vector x is updated by

$$x_{k+1} = x_k - \underbrace{\alpha_k}_{\text{step size (unknown)}} g_k$$

Steepest Gradient Descent

Quadratic optimization:

$$\min_{\mathbf{x} \in \mathbb{R}^2} f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^\top \mathbf{A} \mathbf{x} - \mathbf{b}^\top \mathbf{x}$$

Rewrite the objective function:

$$\Rightarrow f(\mathbf{x}_{k+1}) = f(\mathbf{x}_k - \alpha_k \mathbf{g}_k)$$

Steepest Gradient Descent

Quadratic optimization:

$$\min_{\mathbf{x} \in \mathbb{R}^2} f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^\top \mathbf{A} \mathbf{x} - \mathbf{b}^\top \mathbf{x}$$

Rewrite the objective function:

$$\begin{aligned} \Rightarrow f(\mathbf{x}_{k+1}) &= f(\mathbf{x}_k - \alpha_k \mathbf{g}_k) \\ &= \frac{1}{2} (\mathbf{x}_k - \alpha_k \mathbf{g}_k)^\top \mathbf{A} (\mathbf{x}_k - \alpha_k \mathbf{g}_k) - \mathbf{b}^\top (\mathbf{x}_k - \alpha_k \mathbf{g}_k) \end{aligned}$$

Steepest Gradient Descent

Quadratic optimization:

$$\min_{\mathbf{x} \in \mathbb{R}^2} f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^\top \mathbf{A} \mathbf{x} - \mathbf{b}^\top \mathbf{x}$$

Rewrite the objective function:

$$\begin{aligned} \Rightarrow f(\mathbf{x}_{k+1}) &= f(\mathbf{x}_k - \alpha_k \mathbf{g}_k) \\ &= \frac{1}{2} (\mathbf{x}_k - \alpha_k \mathbf{g}_k)^\top \mathbf{A} (\mathbf{x}_k - \alpha_k \mathbf{g}_k) - \mathbf{b}^\top (\mathbf{x}_k - \alpha_k \mathbf{g}_k) \\ &= \frac{1}{2} \alpha_k^2 \mathbf{g}_k^\top \mathbf{A} \mathbf{g}_k - \alpha_k \mathbf{g}_k^\top \mathbf{A} \mathbf{x}_k + \alpha_k \mathbf{b}^\top \mathbf{g}_k + \underbrace{\frac{1}{2} \mathbf{x}_k^\top \mathbf{A} \mathbf{x}_k - \mathbf{b}^\top \mathbf{x}_k}_{\text{constant}} \end{aligned}$$

Steepest Gradient Descent

Quadratic optimization:

$$\min_{\mathbf{x} \in \mathbb{R}^2} f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^\top \mathbf{A} \mathbf{x} - \mathbf{b}^\top \mathbf{x}$$

Rewrite the objective function:

$$\begin{aligned} \Rightarrow f(\mathbf{x}_{k+1}) &= f(\mathbf{x}_k - \alpha_k \mathbf{g}_k) \\ &= \frac{1}{2} (\mathbf{x}_k - \alpha_k \mathbf{g}_k)^\top \mathbf{A} (\mathbf{x}_k - \alpha_k \mathbf{g}_k) - \mathbf{b}^\top (\mathbf{x}_k - \alpha_k \mathbf{g}_k) \\ &= \frac{1}{2} \alpha_k^2 \mathbf{g}_k^\top \mathbf{A} \mathbf{g}_k - \alpha_k \mathbf{g}_k^\top \mathbf{A} \mathbf{x}_k + \alpha_k \mathbf{b}^\top \mathbf{g}_k + \underbrace{\frac{1}{2} \mathbf{x}_k^\top \mathbf{A} \mathbf{x}_k - \mathbf{b}^\top \mathbf{x}_k}_{\text{constant}} \\ &= \frac{1}{2} \alpha_k^2 \underbrace{\mathbf{g}_k^\top \mathbf{A} \mathbf{g}_k}_{\geq 0} - \alpha_k \mathbf{g}_k^\top \underbrace{(\mathbf{A} \mathbf{x}_k - \mathbf{b})}_{=\mathbf{g}_k} + \text{constant} \end{aligned}$$

Steepest Gradient Descent

Optimize step size α_k :

$$\min_{\alpha_k} p(\alpha_k) = \frac{1}{2} \alpha_k^2 \underbrace{\mathbf{g}_k^\top \mathbf{A} \mathbf{g}_k}_{\geq 0} - \alpha_k \underbrace{\mathbf{g}_k^\top \mathbf{g}_k}_{\geq 0}$$

Steepest Gradient Descent

Optimize step size α_k :

$$\min_{\alpha_k} p(\alpha_k) = \frac{1}{2} \alpha_k^2 \underbrace{\mathbf{g}_k^\top \mathbf{A} \mathbf{g}_k}_{\geq 0} - \alpha_k \underbrace{\mathbf{g}_k^\top \mathbf{g}_k}_{\geq 0}$$

③ The optimal α_k :

$$\frac{dp(\alpha_k)}{d\alpha_k} = 0 \quad (\text{first-order derivative is zero})$$

Steepest Gradient Descent

Optimize step size α_k :

$$\min_{\alpha_k} p(\alpha_k) = \frac{1}{2} \alpha_k^2 \underbrace{\mathbf{g}_k^\top \mathbf{A} \mathbf{g}_k}_{\geq 0} - \alpha_k \underbrace{\mathbf{g}_k^\top \mathbf{g}_k}_{\geq 0}$$

③ The optimal α_k :

$$\frac{dp(\alpha_k)}{d\alpha_k} = 0 \quad (\text{first-order derivative is zero})$$

$$\Rightarrow \alpha_k \mathbf{g}_k^\top \mathbf{A} \mathbf{g}_k - \mathbf{g}_k^\top \mathbf{g}_k = 0$$

$$\Rightarrow \alpha_k = \frac{\mathbf{g}_k^\top \mathbf{g}_k}{\mathbf{g}_k^\top \mathbf{A} \mathbf{g}_k}$$

Steepest Gradient Descent

Quadratic optimization:

$$\min_{\mathbf{x} \in \mathbb{R}^2} f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^\top \mathbf{A} \mathbf{x} - \mathbf{b}^\top \mathbf{x}$$

△ The update equations:

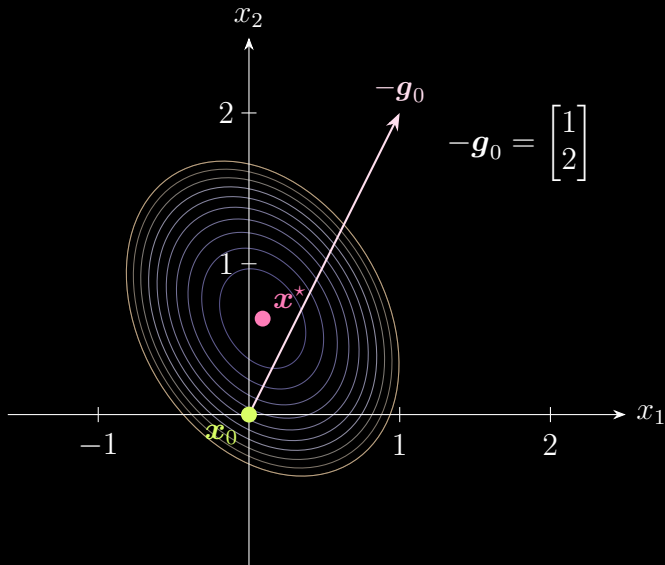
$$\mathbf{g}_k = \mathbf{A} \mathbf{x}_k - \mathbf{b} \quad (\text{gradient})$$

$$\alpha_k = \frac{\mathbf{g}_k^\top \mathbf{g}_k}{\mathbf{g}_k^\top \mathbf{A} \mathbf{g}_k} \quad (\text{step size})$$

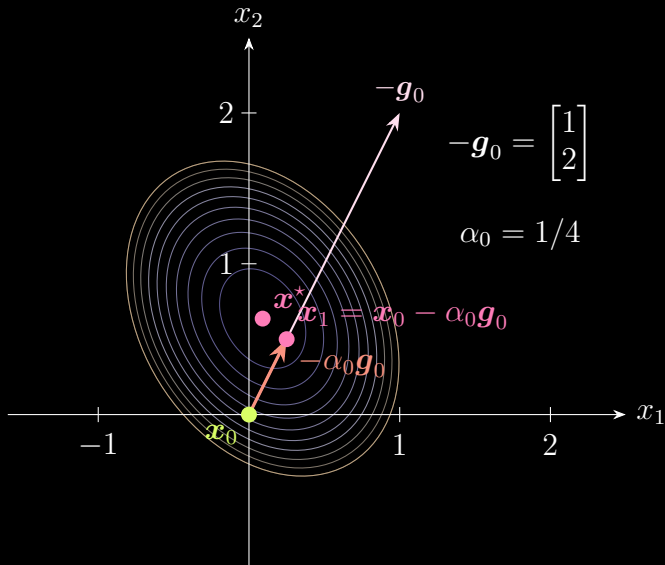
$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha_k \mathbf{g}_k \quad (\text{decision vector})$$

with an initial point $\mathbf{x}_0$.

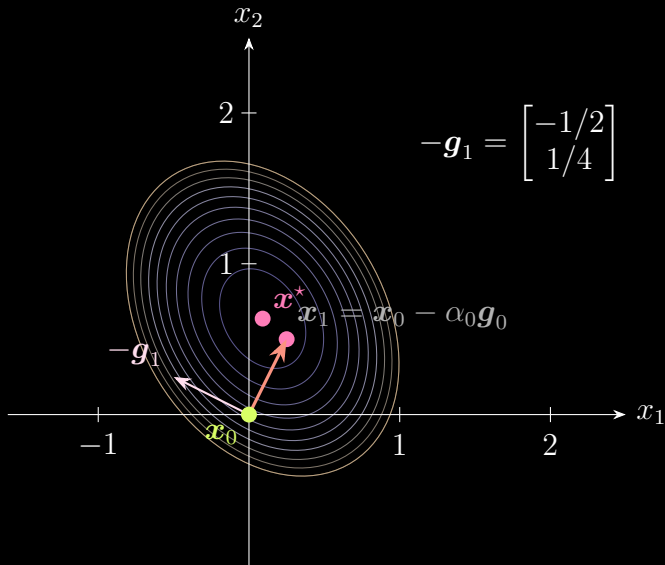
Steepest Gradient Descent for Quadratic Optimization



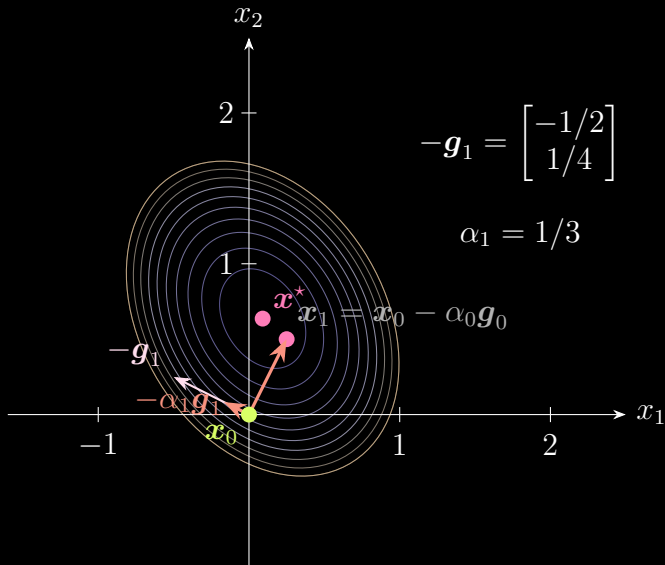
Steepest Gradient Descent for Quadratic Optimization



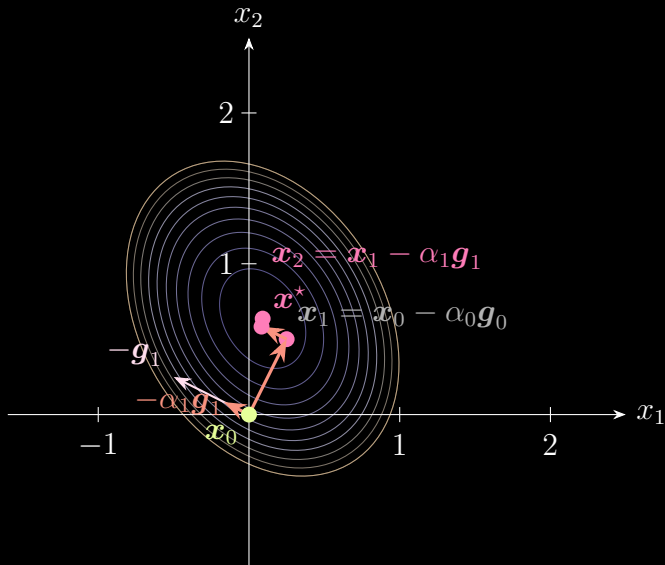
Steepest Gradient Descent for Quadratic Optimization



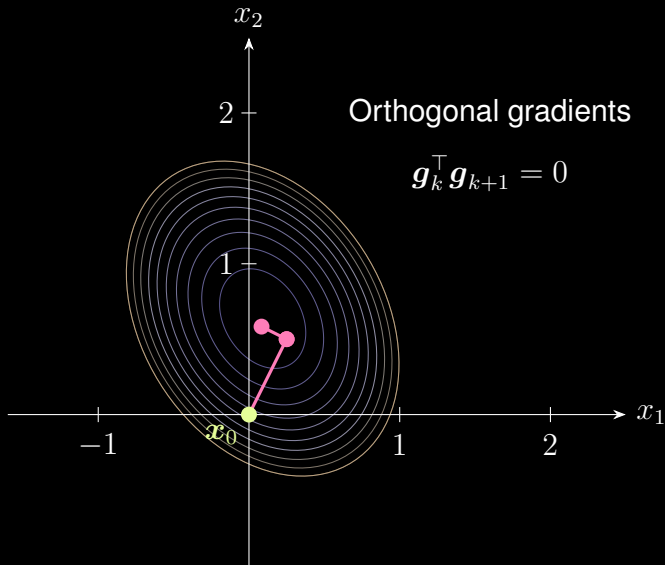
Steepest Gradient Descent for Quadratic Optimization



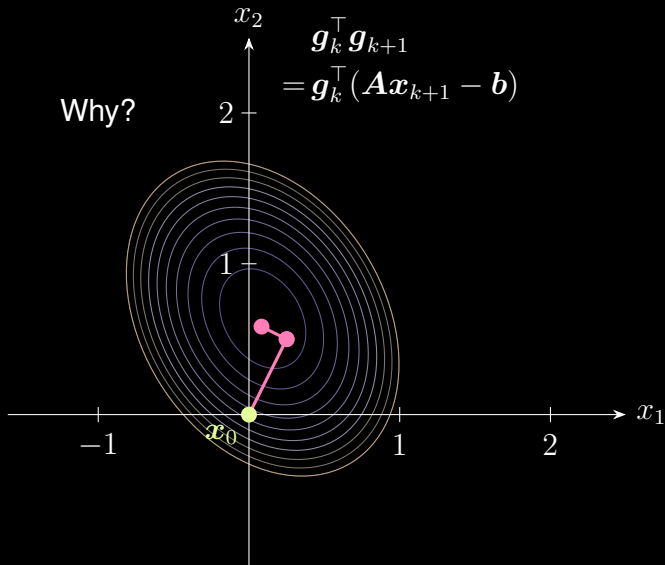
Steepest Gradient Descent for Quadratic Optimization



Steepest Gradient Descent for Quadratic Optimization

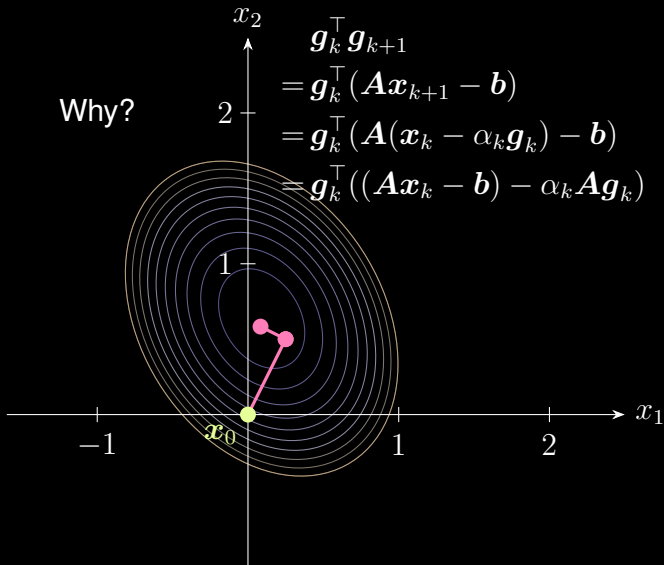


Steepest Gradient Descent for Quadratic Optimization



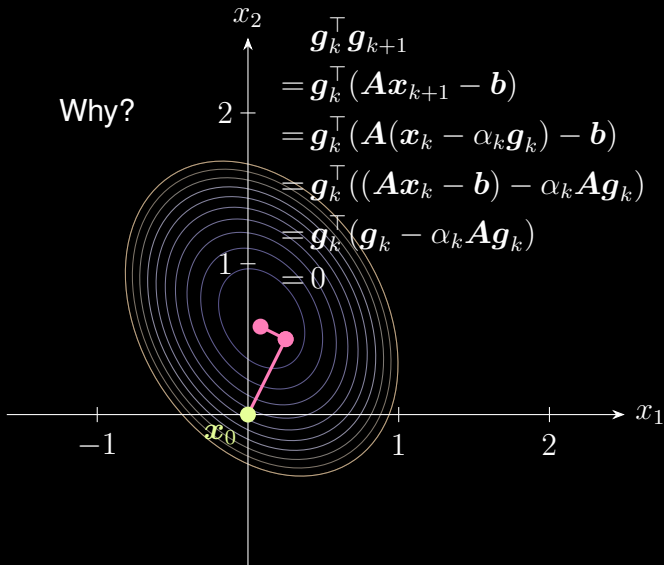
Steepest Gradient Descent for Quadratic Optimization

Why?

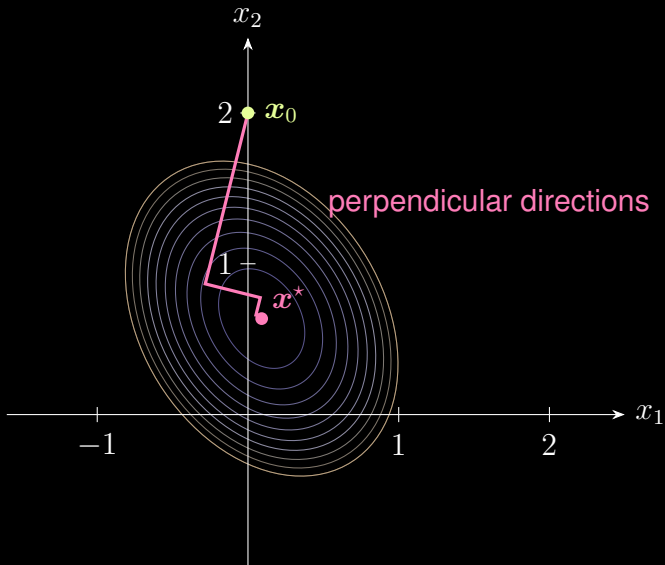


Steepest Gradient Descent for Quadratic Optimization

Why?

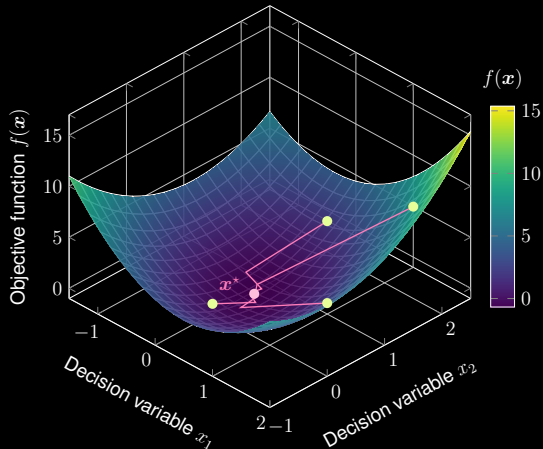
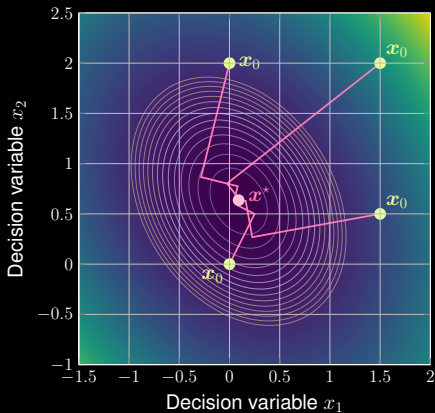


Steepest Gradient Descent for Quadratic Optimization



Steepest Gradient Descent for Quadratic Optimization

$$\min_{\mathbf{x} \in \mathbb{R}^2} \frac{1}{2} \mathbf{x}^\top \mathbf{A} \mathbf{x} - \mathbf{b}^\top \mathbf{x} \quad \text{with} \quad \mathbf{A} = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$



Thanks for your attention!

About me:

 Homepage: <https://xinychen.github.io>

 How to reach me: chenxy346@gmail.com